

Subscale Wind Tunnel Experiments on the Aerodynamics of a Rotor Approaching the Landing Deck of a Generic Destroyer

Wei-Han Chen
Doctoral Candidate

Dounya Jarrad
Graduate Research Assistant

Juergen Rauleder
Associate Professor

Daniel Guggenheim School of Aerospace Engineering
Georgia Institute of Technology
Atlanta, GA 30332, USA

ABSTRACT

This study presents subscale wind tunnel experiments investigating the transient aerodynamic interactions of a rotor during a continuous relative-wind approach toward the landing deck of the NATO Generic Destroyer. Time-resolved rotor loads and stereoscopic particle image velocimetry measurements were used to characterize the interacting ship-rotor flow field under headwind and quartering wind-over-deck conditions. The measurements captured the evolving influence of ship airwake, ground effect, and superstructure-induced recirculation as the rotor moved from downstream to the final hover position over the deck. The results show that rotor thrust, rolling moment, and pitching moment underwent distinct changes throughout the approach, with the loading trends varying significantly with wind-over-deck angle. Time-frequency analysis further reveals that the unsteady rotor response was concentrated in a limited band of frequencies associated with various coherent flow structures shed from the ship superstructure. Spectral proper orthogonal decomposition was used to identify the dominant airwake features responsible for these fluctuations at specific frequencies, including large-scale structures originating from the radar and hangar region. These findings demonstrate that the dynamic approach resolved both slow- and fast-changing transient aerodynamic effects along the approach path that cannot be captured with static hovering measurements.

NOMENCLATURE

C_T	Thrust coefficient, $\frac{T}{\rho(\pi R^2)(\Omega R)^2}$	c	Blade chord (m)
C_{M_x}	Rolling moment coefficient, $\frac{M_x}{\rho(\pi R^2)(\Omega R)^2 R}$	f	Frequency (Hz)
C_{M_y}	Pitching moment coefficient, $\frac{M_y}{\rho(\pi R^2)(\Omega R)^2 R}$	h_{hod}	Height at hover-over-deck (m)
M_x, M_y	Rotor rolling and pitching moment (N·cm)	i	Mode number
M_{tip}	Rotor blade tip Mach number	l_d	Length of landing deck (m)
N	Number of independent test trials	u, v, w	Flow velocities in x, y, and z directions (m/s)
N_b	Number of rotor blades	$\tilde{w}$	Normalized vertical velocity, $\frac{w}{U_\infty}$
R	Rotor radius (m)	w_s, l_s, h_s	Width, length, and height of ship (m)
Re_s	Ship beam Reynolds number, $\frac{U_\infty w_s}{\nu}$	x, y, z	Longitudinal, lateral, and vertical positions relative to center of ship deck (m)
Re_{tip}	Rotor blade tip Reynolds number, $\frac{U_{tip} c}{\nu}$	x_r, y_r, z_r	Longitudinal, lateral, and vertical positions relative to rotor hub center (m)
St	Strouhal number, $\frac{f w_s}{U_\infty}$	$\lambda_i^{(f)}$	Eigenvalue of mode i at frequency f
T	Thrust (N)	λ_L	Length scale between model and full scale
U_∞	Freestream velocity (m/s)	λ_V	Velocity scale between model and full scale
V	Velocity magnitude, $\sqrt{u^2 + v^2 + w^2}$ (m/s)	λ_f	Frequency scale between model and full scale
$\tilde{V}$	Normalized velocity magnitude, $\frac{V}{U_\infty}$	θ_{tw}	Blade twist angle (deg)
V_{app}	Rotor approach speed (m/s)	μ	Advance ratio, $\frac{U_\infty + V_{app}}{\Omega R}$
		ρ	Air density (kg/m ³)
		Ω	Rotor angular velocity (rad/s)
		σ	Rotor solidity, $\frac{N_b c}{\pi R}$
		Ψ	Wind-over-deck angle (deg)
		ν	Kinematic viscosity (m ² /s)

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INTRODUCTION

Helicopter operations around vessels at sea, referred to as the ship-rotor dynamic interface (DI), are governed by the aerodynamic interactions of the unsteady rotor and ship airwakes. These conditions can cause high pilot effort requirements and reduced handling qualities, posing significant risks to pilots and deck operators. It is thus crucial that ship-helicopter operational limits (SHOL) are thoroughly established through the characterization of the aerodynamic interactions in the region, prior to high-risk flight testing and operations.

Various representative ship geometries, examples as shown in Fig. 1, have been developed and utilized alongside existing operational vessels to study the ship-rotor DI problem as a collaborative effort. These include the early Simplified Frigate Shape (SFS) (Ref. 1), its aerodynamically improved successor, SFS-2 (Ref. 2), and the more recently developed NATO Generic Destroyer (NATO-GD) (Ref. 3), which more closely resembles a modern naval destroyer.

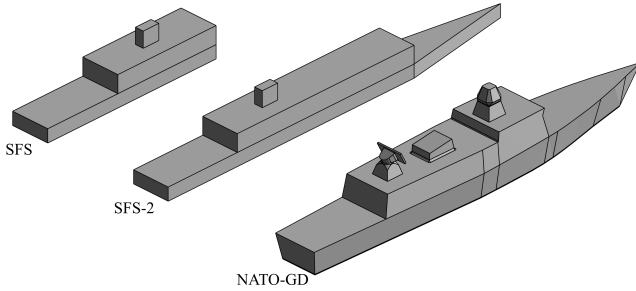


Figure 1: Representative ship geometries.

Extensive experimental studies have been conducted to establish baseline isolated ship airwake behavior. Studies have characterized various ship geometries under the influence of a simulated atmospheric boundary layer (ABL) in scaled wind tunnel tests and analyzed particle image velocimetry (PIV) measurements using proper orthogonal decomposition (POD) to identify the bistable behavior of the turbulent structures around the flight deck (Refs. 4–6). Yon and Li (Ref. 7) recently used PIV and POD to characterize velocity fluctuations around the ship deck and assessed pilot workload factors due to streamwise sinusoidal freestream gusts.

Going beyond the investigation of the ship airwake, researchers have looked into the full DI problem through the inclusion of rotors and full vehicle models in hover surveys and deck approaches. Zan (Ref. 8) conducted one of the earlier experimental studies of the ship-helicopter problem. Time-averaged thrust of a 1:50-scale generic helicopter rotor was measured for a survey of hover heights and positions around the deck of an equivalently geometrically scaled Canadian Patrol Frigate (CPF) model. It was found that there was an overall reduction in thrust within the region with significant thrust gradients across deck positions and between wind conditions.

Approach trajectories have been experimentally evaluated through the discretization of dynamic approaches into quasi-static step sequences of positions along the original approach trajectories, with steady hover measurements taken at each point. Several studies (Refs. 9–12) compared unsteady and time-averaged total rotor loads and PIV flow fields along different approach paths under various wind conditions. Silva et al. (Ref. 13) conducted a comprehensive survey of approach types of 1:48 scale V-22 and H-1 vehicle models to the landing deck of an LPD-17 ship under various wind conditions, finding degradations in combined and individual rotor performance with a high dependency on approach/hover heights and approach directions. However, these quasi-static approach sequences present a limitation in that transient unsteady effects cannot be fully resolved, as the rotor is allowed to reach a steady state at each static location. Only somewhat recently have fully dynamic rotor approaches/motions been experimentally evaluated for the DI problem.

Taymourtash et al. (Ref. 14) compared both quasi-static and fully dynamic stern approaches for a 1:12.5 SFS ship model and 1:10 scale Bo105 rotor model under quartering winds. Beginning with the quasi-static approach cases, the rotor was re-trimmed to a constant thrust and near-zero moments at each discrete location, and these control inputs were then applied along the dynamic approach. Higher unsteady thrust was observed closer to the ship for all cases; however, the quasi-static approach yielded a lower weighted average of unsteady thrust compared to the dynamic approach under quartering wind conditions.

Prior work by Chen and Rauleder (Ref. 15) has touched on the dynamic approach problem in wind tunnel experiments with a generic rotor in a streamwise approach towards the landing deck of a 1:100-scale NATO-GD under headwinds. Time histories of deck surface pressures, rotor loads, and vertical PIV flow fields, as the rotor moved further onto the deck, showed amplified recirculation between the rotor and hangar and an initial transient increase in thrust due to the continuous development of ground effect.

Computationally, Ashok and Rauleder (Ref. 16) developed a mid-fidelity GPU-accelerated lattice-Boltzmann method (LBM) flow field solver and validated the results against new experimental data. Additionally, two-way coupled real-time simulations were achieved in a full-scale ship-rotorcraft dynamic approach scenario, which enabled the possibility for the LBM flow field solver to be integrated into real-time piloted simulations.

In the related realm of experimental rotor aerodynamics with dynamic rotor motions, recent experimental work by Semelka and Rauleder (Ref. 17) characterized the transient interactions of a proprotor-wing configuration during a dynamic tiltrotor conversion maneuver through PIV and wing load/pressure measurements. Several challenges were identified during testing, including the changing blockage effect caused by moving structures and the propeller energizing the freestream flow at lower tilt angles. Through a novel interpolating inference scheme, larger deviations in wing loads from the

quasi-static baseline were observed for faster conversions, along with differences in transient wing lift between opposing conversion directions.

The challenges of experimenting with rotors in dynamic motion have limited the amount of publicly available data for such conditions in wind tunnels, especially related to the ship-rotor DI problem. Therefore, the present study seeks to resolve the transient load development and ship-rotor interaction mechanisms during a continuous approach toward the landing deck of a generic destroyer under multiple wind-over-deck (WOD) angles, with particular emphasis on the influence of ground effect, recirculation, and coherent unsteady flow features on rotor performance. By combining rotor load measurements with time-resolved stereoscopic PIV and frequency-domain analysis, this work aims to improve the physical understanding needed to support the establishment of SHOLs and to inform future experimental and modeling efforts in shipboard recovery scenarios.

METHODOLOGY

Experimental Facility

Experiments were conducted in the John J. Harper Wind Tunnel at the Georgia Institute of Technology. The closed-return tunnel has an optically accessible rectangular test section, measuring 2.34 m (7.67 ft) in width, 2.13 m (7 ft) in height, and 3.22 m (10.58 ft) in length. The same experimental setup as in (Ref. 18) was used, shown in Fig. 2, and consists of a rigid platform spanning the length and width of the test section, elevated 0.7 m (2.3 ft) above the test section floor. The rotor model was positioned around the ship using a three-axis servo-motor-driven traverse system located on the platform downstream of the ship model. Linear encoders were mounted along all three axes of the traverse for precise positioning and tracking of the rotor. The origin, or zero position, of the rotor was established by carefully aligning the rotor with the hover-over-deck (HOD) position above the center of the landing deck.

Models and Instrumentation

The ship and rotor models used in this study were 1:100-scale versions of the NATO-GD and NATO Generic Rotorcraft



Figure 2: Experimental setup in the Harper Wind Tunnel.

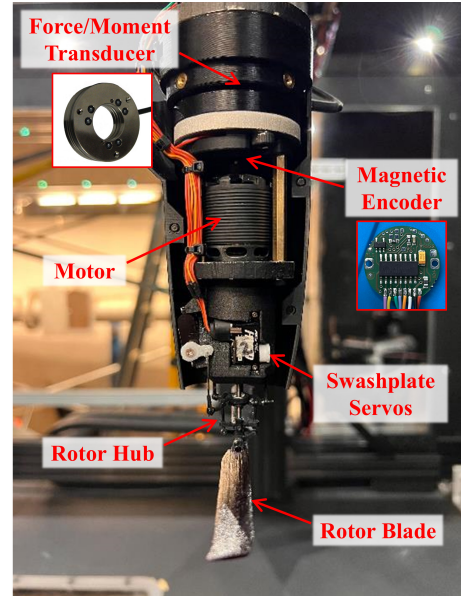


Figure 3: Assembly of the rotor model.

(NATO-GR), respectively. The NATO-GR rotor, which is based on the Sikorsky SH-60 Seahawk, was developed in conjunction with the NATO-GD. The rotor blade geometry incorporates an untapered planform and swept tip, uniform NACA 0012 airfoil sections, and a total geometric twist of $\theta_{tw} = -12.5$ deg. The detailed geometries of the rotor and ship models were acquired from the repository provided by the NRC Canada (Ref. 3) and the NATO AVT-315 task group (Ref. 19).

The ship model, shown in the center of Fig. 2, was assembled from 3D-printed sections and mounted on the platform, and could be yawed about the center of the landing deck in increments of 5 deg to a maximum of 60 deg in either direction. Although not discussed in this present study, ship deck pressures were also measured using a total of 126 evenly distributed pressure taps: 98 on the landing deck surface and 28 on the hangar wall. Pressures from each tap were recorded through two 64-channel differential pressure scanners housed within the ship model. A pitot-static probe, located upstream of the platform and laterally offset from the ship, was used to measure the freestream conditions.

The rotor model (Fig. 3) consists of a brushless motor driving a two-bladed hingeless hub in a counter-clockwise direction with a swashplate assembly sourced from a commercial off-the-shelf radio-controlled helicopter. Three micro servos were used to actuate both collective and cyclic pitch controls of the swashplate. The rotor blades were fabricated in-house with carbon fiber and epoxy resin, reproducing the planform and twist profile of the NATO-GR rotor blades. However, since a two-bladed rotor hub assembly was used in this study, the rotor geometry used was a slight modification of the original four-bladed design of the NATO-GR, where the chord length of the test model was twice the length of a directly scaled NATO-GR, such that the rotor solidity (σ) was maintained.

The rotor azimuth position and rotational speed were tracked using a magnetic rotary encoder (8192 counts/rev) mounted

behind the motor, and hub loads were measured using a six-degree-of-freedom force/torque transducer (Nano43, ATI Industrial Automation) with a sampling rate of 12 kHz. The target rotor angular speed (Ω) and rotor trim inputs were maintained through proportional-integral-derivative (PID) controllers using feedback from the encoder and force/torque transducer, respectively. The entire assembly was encased in an aerodynamic fairing and attached to the central vertical arm of the traverse. Further details on the hardware and data acquisition system were elaborated on and can be found in the prior work presented in the 81st annual forum (Ref. 18).

Model Scaling Specifications

The scaled testing conditions of the experiment are listed alongside the full scale reference condition in Table 1. The rotor radius was directly geometrically scaled, but, as mentioned previously, due to the reduced blade count, a wider blade chord length (c) was used to match the full-scale rotor solidity (σ). The test conditions were primarily determined by the following scaling parameters: the rotor advance ratio (μ) and thrust coefficient (C_T), and the Reynolds number based on the ship beam width (Re_s). The μ here consists of the contributions from both the freestream (U_∞) and the rotor approach speed (V_{app}). The rotor C_T was set based on a vehicle gross weight of 9000 kg (19,842 lbs).

However, directly matching Re_s in such small-scale testing conditions is difficult due to the substantial increase in the corresponding freestream velocity (U_∞) required. Therefore, a compromise was made by only matching μ and C_T of the rotor, while maintaining sufficiently high Re_s to ensure Reynolds number independence of the ship airwake characteristics. Several experimental studies (Refs. 6, 20, 21) have shown that the ship airwake is insensitive to Reynolds number above

a range of 10^4 to 10^5 of the ship beam Reynolds number. Additionally, tests conducted in-house for the same 1:100 NATO-GD ship model in isolation confirmed no significant differences in velocity profile and turbulence characteristics above the landing deck from 30 m/s ($Re_s \approx 4.6 \times 10^5$) down to 3 m/s ($Re_s \approx 4.6 \times 10^4$). Thus, although the ship beam Reynolds number was not directly matched, similarity of the ship airwake characteristics to those in high- Re or full-scale conditions was expected. Nonetheless, it should be noted that the blade tip Mach number (M_{tip}) and the blade tip Reynolds number (Re_{tip}) could not be matched to the full-scale reference due to limitations in the structural integrity of the rotor hub when operating at the high angular speed required.

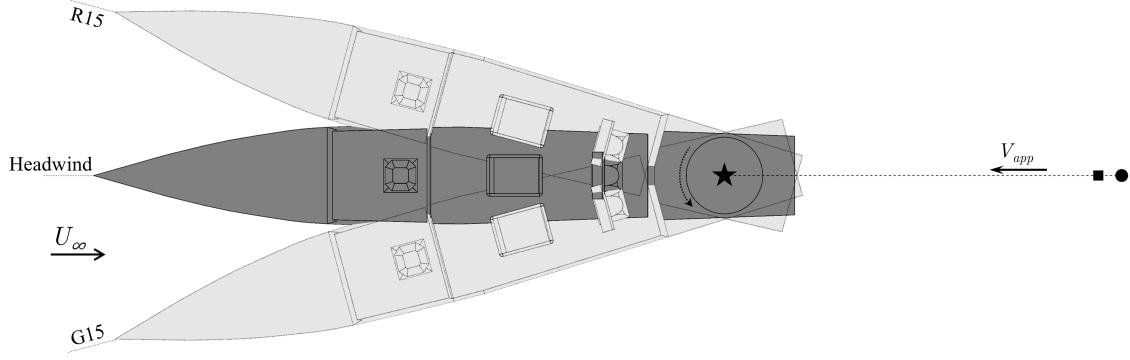
Approach Trajectory and Testing Procedures

The dynamic approach paths evaluated in this study are relative-wind approaches under various WOD angles. As defined in the top view in Fig. 4a, for a given WOD angle, the relative-wind approach path, shown by the dashed line, tracks along the freestream direction, approaching directly upstream towards the HOD position marked by the star symbol. Three WOD angles considered in this study: 15 deg red winds (R15), where the ship yawed right towards the starboard direction, 15 deg green winds (G15), where the ship yawed left towards the port-side direction, and a baseline 0 deg headwind case. Additionally, the circle on the ship deck shows the relative size of the rotor disk, with the dashed arrow indicating the rotation direction of the rotor. These approaches followed a 3 deg glide slope, as shown in the side view in Fig. 4b. A fixed approach speed relative to the ship (V_{app}) of 0.025 m/s was used, which was limited by the capability of the traverse system. This speed was dynamically equivalent to approximately 0.2 knots (0.1 m/s) at full-scale. This velocity represents a rather slow approach, but future tests are underway with an improved traverse system to explore higher approach speeds.

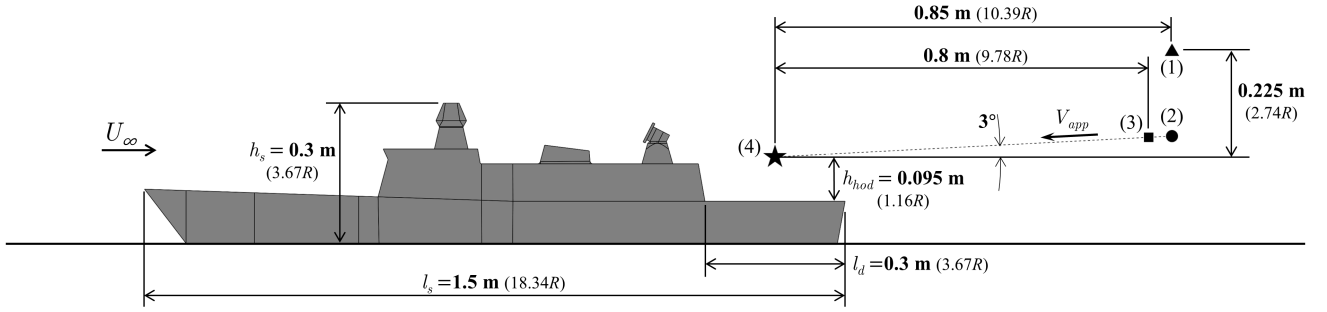
Before executing the approach, the rotor was first automatically trimmed to the target thrust and zero in-plane moments at the up-and-away position, indicated by the triangle symbol at position (1) in Fig. 4b, outside of the influence of the ship airwake. Trim loads at this location were recorded for 10 seconds. This control input was then kept fixed throughout each run. Next, the rotor was brought directly down to position (2), where it intersected the 3-deg glide slope. After the loads were stabilized, the approach sequence was initiated. The rotor was accelerated to the approach speed over a 1-second period before data acquisition was triggered at position (3), marked by the square symbol 0.8 m downstream of the HOD position. A constant approach speed was maintained while traversing the rotor towards the HOD position, marked as the star symbol at location (4). A 1-second deceleration period was used to bring the rotor to a stop at the HOD position. The rotor was held at the final HOD position for 10 seconds after completing the approach to capture the transient fluctuations. Hub loads and rotor position data were continuously recorded throughout the approach and final hover segment at a sampling rate of 12 kHz. Once the full approach and hover sequence was complete, the

Table 1: Testing conditions and scaling parameters.

	Reference Condition at Full Scale	Wind Tunnel Model Scale
Ship		
U_∞	20.6 m/s (40 kts)	5 m/s
l_s	150 m	1.5 m
w_s	22.1 m	0.221 m
h_s	30.3 m	0.303 m
Re_s	3.1×10^7	7.6×10^4
Rotor		
R	8.2 m	0.082 m
N_b	4	2
c	0.527 m	0.010 m
Ω	27.0 rad/s (258 RPM)	656.5 rad/s (6269 RPM)
M_{tip}	0.64	0.16
Re_{tip}	8.0×10^6	3.9×10^4
σ		0.08
μ		0.094
C_T	0.0075 (Out of ground effect)	
V_{app}	0.2 kts (0.1 m/s)	0.025 m/s



(a) Top view



(b) Side view

Figure 4: Relative positions and the approach trajectory.

rotor was traversed back to the up-and-away position, and loads were recorded for another 10 seconds to ensure consistency in the initial trim load measurements. A total of six repeatability runs were done for each WOD angle, including three runs that were completed with PIV measurements.

Particle Image Velocimetry

Stereoscopic PIV was used to evaluate the flow fields around the rotor and landing deck during final approach, in addition to the rotor hub loads. Hardware setup was identical to that shown in the prior work by Chen and Rauleder (Ref. 18). Imaging was conducted for various vertical planes oriented parallel to the approach path of the rotor. These included a

vertical plane that passed through the center of the rotor hub and two additional planes offset $\pm 0.5 R$ (± 0.04 m) from the center plane to capture the flow field around the advancing and retreating sides of the rotor, respectively. Images were captured at a frequency of 100 Hz for a total of 20 seconds, with 15 seconds allocated to the final approach and 5 seconds to capture the rotor hovering at the HOD position.

Laser beams were generated using a Nd:YLF dual-cavity pulsed laser (Photonics DM30-527-DH, 527 nm, 30 mJ/pulse) with a pulse separation time of 150 μ s. In order to ensure the entire approach path would be illuminated and not eclipsed by the traverse arm, the beams were split into two using a 50:50 beam splitter and rerouted to illuminate the upstream and downstream sides of the rotor and deck, as shown in Fig. 5. Finally, the emitted beams were passed through adjustable sheet optics containing cylindrical lenses with focal length $f = -20$ mm, and expanded into 2 mm-thick laser sheets. The routing was finely tuned to ensure the two laser sheets were coplanar. These sheet optics were mounted on individual traverses to allow for automatic and precise lateral positioning of the combined laser plane.

A Rosco Vapour fog generator with propylene glycol fog fluid was used to produce seeding particles with a nominal diameter of 4 μ m. The particles were released downstream of the test section, and the density was carefully tuned prior to the approach maneuvers to ensure uniform seeding throughout the test section in each run. Two Phantom high-speed cameras

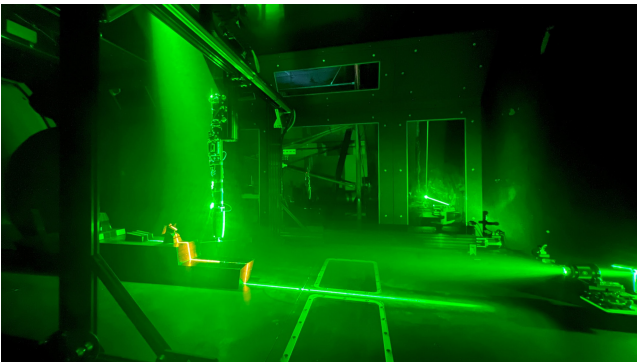


Figure 5: Laser sheets illuminating the landing deck.

(2560×1600 pixels full resolution, $10 \mu\text{m}$ pixel size), fitted with Scheimpflug adapters and 100 mm lenses with an aperture setting of $T/3.1$, were mounted outside of the test section on either side of the vertical laser planes, facing upstream towards the landing deck region. To reduce surface reflections from the models in the high-speed images, the hangar, deck, and adjacent surfaces were painted with 3 coats of a custom 6 mg/mL mixture of clear gloss acrylic paint and fluorescent Sulforhodamine B (Kiton Red 620, KR620) dye. The dissolved dye reemitted the absorbed laser light at a longer wavelength of $\sim 580 \text{ nm}$, as seen in the orange region on the ship model in Fig. 5, which was filtered out through $527 \pm 10 \text{ nm}$ bandpass filters mounted on the camera lenses.

The acquired image pairs were processed using DaVis 10.2 software by LaVision Inc. Processing began with background removal to eliminate both static and transient reflections from the models and the rotating blades. The background-subtracted images were then analyzed using a standard multi-pass cross-correlation method, starting with an interrogation window of 64×64 pixels at 50% overlap and concluding with a final window of 32×32 pixels at 75% overlap. A universal outlier detection scheme (Ref. 22) with a 5×5 vector filter was applied to iteratively identify and remove spurious velocity vectors. Regions obstructed by the ship, rotor, or intense light reflections were masked and excluded from vector calculation. As a result, the final vector spacing varied slightly among measurement planes, depending on the laser sheet position relative to the cameras, with an average spacing of approximately 1.3 mm.

MACROSCOPIC EVOLUTION OF ROTOR LOADS AND FLOW FIELDS

In this section, the overall trends in rotor loads and flow fields around the ship's landing deck are examined as the rotor approached the ship. To obtain these general trends, moving averages with a Gaussian windowing function of 0.5 seconds were applied to all measurement data to filter out most of the large variations caused by highly oscillatory, rotor-rotation-induced aerodynamic and vibrational loads, as well as high-frequency unsteady features within the flow fields. This helps better visualize the changes associated with the rotor's proximity to the ship as it approaches the deck.

Compensation of Inertial Effects

Although the majority of the dynamic approach processes were conducted at a constant approach speed, there were also phases where the rotor experienced acceleration and deceleration: mainly the initial startup from the starting point and the transition into hover at HOD. These phases induced additional inertial forces and moments caused by the mass of the rotor support structure, which were also captured by the force transducer. In particular, the deceleration while approaching the HOD had the most significant impact on the results and had to be corrected. To compensate for this, a set of load measurements was conducted along each approach

trajectory under quiescent flow ($U_\infty = 0 \text{ m/s}$) without the rotor spinning ($\Omega = 0 \text{ RPM}$). This essentially captured only the inertial loads resulting from accelerations, decelerations, and inherent vibrations caused by the traverse movement. These time series measurements, averaged across four trial runs, were then subtracted from all subsequent experiments along the same approach path.

An example can be seen in Fig. 6, which is a snippet of an approach around when the rotor transitioned into hover at the HOD ($t = 0$). Negative timestamps indicate segments before reaching the hover position. Figure 6a shows the approach speed both commanded and from the integrated encoders on the traverse system. There was a small increase in velocity right at the end of deceleration, mainly due to a slightly mistuned feedback controller, in which the integral term boosted the motors to compensate for the remaining small positional error.

Figure 6b plot shows the variation in the rotor pitching moment, which was the most severely impacted by the inertial effects due to the direction of deceleration and the large distance between the center of mass of the rotor assembly and the moment reference center. Again, note that the shown loads were moving averages with Gaussian windowing to filter out the large oscillations from the rotor rotation. The dotted line shows the measured rotor pitching moment during the quiescent approach without the rotor operating. This was ensemble-averaged over four trial runs, with 95% confidence intervals marked by the shaded gray region.

Next, the original uncorrected rotor pitching moment from one

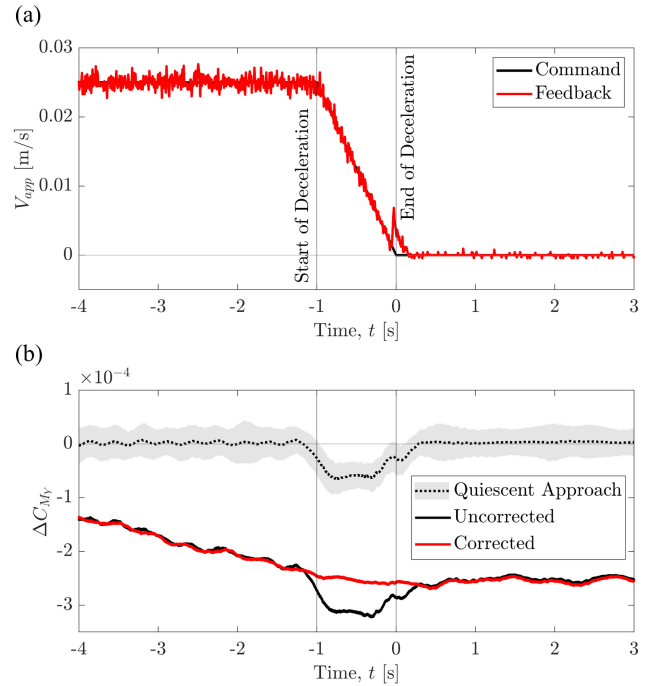


Figure 6: Example time series of the (a) rotor approach speed and (b) the pitching moment contaminated by inertial loads with corrected results.

of the actual tests, shown in a black solid line, also showed a noticeable decrease during deceleration that largely resembles the pattern observed in the quiescent approach. This directly illustrates the influence of inertia during the deceleration period, which appeared as an additional pitch-down effect about the rotor hub. The maximum deviation in the pitching moment seen in Fig. 6b caused by the inertia of the structure was more than 26 percent ($\Delta C_{My} \approx 7.5 \times 10^{-6}$) of the true values. After subtracting the inertial effect from the original data, i.e., subtracting the dashed line from the black solid line, the trend in pitching moment appeared to become smooth during the transition into hover, as shown by the red line. This signifies that the correction for inertial effects was necessary in order to obtain high-quality results and avoid misinterpretation in experiments involving dynamic motion.

Wind-Over-Deck Angle Effects on Rotor Loads

To assess how WOD angles affect the rotor performance throughout the approach towards the landing deck, the measured rotor hub loads were examined. Figure 7 shows the delta in rotor hub loads relative to the trimmed up-and-away measurements along the relative-wind approach path under different WOD angles. Each line represents the

ensemble-averaged time history across six independent test trials, filtered using a Gaussian-weighted moving average as described in the previous section. Additionally, two horizontal axes were plotted for the ease of comparison. The bottom axis indicates time after reaching the hover position. Again, negative timestamps indicate times before reaching the hover position. The top axis with black vertical grid lines indicates the longitudinal position of the rotor (x) downstream of the HOD, normalized by the length of the ship deck (l_d). For example, a x/l_d of 0 corresponds to the rotor at the HOD position, and 0.5 is when the center of the rotor disk was directly above the stern of the ship. The frame of reference for the rotor loads had the moment reference center at the rotor hub and follows the convention in which the X -axis points in the direction of the rotor heading; in this case, directly upstream into the freestream wind. The Y -axis then points to the right (starboard) side of the vehicle.

Figure 7a shows the delta in rotor thrust coefficients (ΔC_T) along the approach path. The loss in rotor thrust under certain conditions has previously been shown to be mainly due to the loss in translational lift on the advancing side of the rotor when entering the low-velocity ship airwake region (Ref. 15). At the beginning of the approach, the rotor under a headwind and a G15 WOD experienced a similar level of thrust reduction from the developed ship airwake. However, under a R15 WOD, the rotor produced significantly less thrust due to the influence of the ship airwake being skewed towards the advancing side of the rotor. This was most likely a combined effect of the reduced freestream within the ship airwake on the advancing side of the rotor and the less-influenced (faster) freestream on the retreating side, resulting in an overall decrease in thrust across the rotor disk. These adverse effects strengthened as the rotor moved closer to the landing deck in both the headwind and R15 WOD conditions. On the other hand, a G15 WOD showed increased beneficial effects as the rotor approached the ship deck. Due to the less dissipated ship airwake near the deck, the rotor experienced a more pronounced asymmetric airwake impingement, which resulted in an opposite effect from that under a R15 WOD. The advancing side saw a relatively undisturbed freestream, and the retreating side was subjected to the low-speed ship airwake, collectively inducing a slight increase in rotor thrust.

Interestingly, the delta in hub rolling moments (ΔC_{Mx}), shown in Fig. 7b, exhibited similar trends across all WOD conditions, with a near-neutral change starting furthest downstream of the ship. As the distances between the ship and rotor were reduced along the approach, the rolling moments gradually increased in all cases, resulting in a slight positive (right) rolling moment while hovering above the landing deck. This suggests that, although influenced by the asymmetric ship airwake under different WOD angles, the thrust changes across the lateral direction of the rotor canceled each other out along the relative-wind approach path.

Lastly, Fig. 7c shows the change in the pitching moments about the rotor hub (ΔC_{My}) along the approach. Both the headwind and R15 WOD cases started with similar positive (pitch-up) moments, while the G15 WOD started with a slightly negative

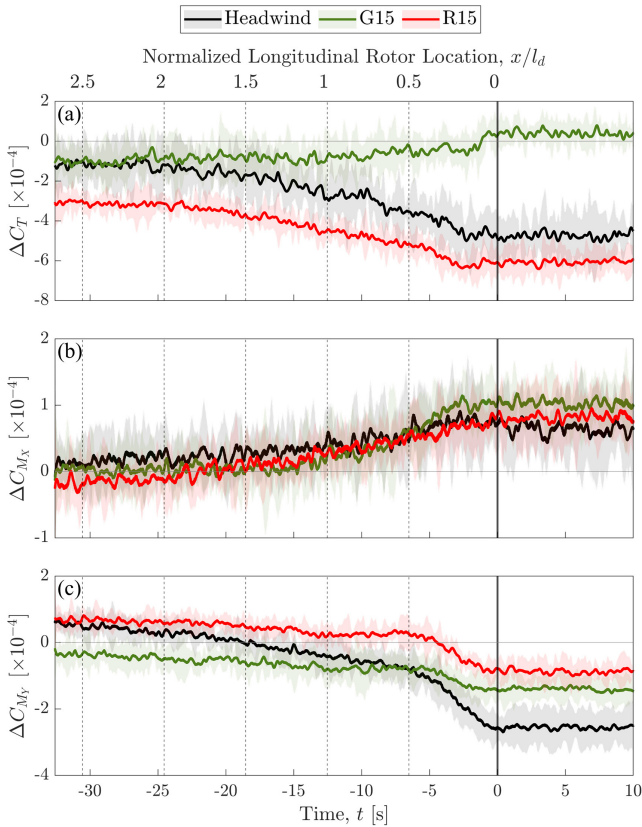


Figure 7: Change in rotor hub loads relative to up-and-away, during a relative-wind approach under different WOD angles. Shaded regions indicate 95% confidence intervals ($N = 6$).

moment. The rotor under a headwind then experienced a gradual decrease in pitching moment, while the G15 and R15 WOD remained relatively constant along the approach paths. It wasn't until the rotor hub crossed the stern ($x/l_d = 0.5$) and entered the landing deck that it was subjected to a rapid pitch-down tendency. During this final approach stage, the rotor entered a region with significantly lower freestream velocity due to the ship airwake, reducing the effective rotor advance ratio and hence decreasing the fore-aft bias in the rotor inflow. This effect was further enhanced by the ground effect and separated flow behind the superstructure. A similar observation regarding ground effect was also reported by Curtiss et al. (Ref. 23), where a reduction in pitching moment was seen as the height above ground was reduced with a rotor in forward flight. From Fig. 7c, this pitch-down tendency was most pronounced under headwind and least significant under a G15 WOD. This revealed that, although the changes in pitching moment were mainly related to the fore-aft thrust imbalance on the rotor, the contributions of the advancing and retreating sides of the rotor were significantly different under different quartering winds.

Flow Fields During Final Approach

To support the trends observed in the integrated rotor hub loads, specifically the rapid pitch-down near the ship deck, flow fields from the PIV experiments were analyzed. Since the goal is to study slowly changing features, the flow fields shown here were also averaged and filtered using the same Gaussian windowing function of 0.5 seconds as the previously shown rotor load data. With a sampling rate of 100 Hz, this resulted

in an averaging window of 51 flow field snapshots centered at each time instance, with 25 snapshots before and 25 snapshots after.

Firstly, to examine the strong and sudden pitch-down tendency discussed earlier as the rotor approached the ship deck under headwind, flow fields around the rotor were extracted at four different locations equally spaced between the stern and the final HOD position, as shown in Fig. 8. The first row shows the velocity magnitudes ($\tilde{V}$) and streamlines of the flow field, and the second row shows the delta in vertical velocities ($\Delta\tilde{w}$) between an isolated rotor at each location along the final approach. Note that the isolated rotor flow fields were obtained separately from the dynamic approach tests and were conducted with a rotor hovering under the same freestream velocity and out of ground effect to reflect conditions that were similar to the up-and-away. It was particularly challenging to obtain PIV at the up-and-away position; thus, a separate test was conducted using an isolated rotor without the ship as a reference condition. This comparison was done to better visualize changes in flow features relative to the trim condition.

By looking at the velocity magnitudes in the top row in Fig. 8, column (d) shows that although the rotor was directly above the stern, meaning the rotor disk was halfway onto the ship deck, ground effect was not yet noticeable due to the backward-skewed wake. As the rotor moved inwards onto the deck, at the position shown in column (c), the rotor wake from the upstream half of the rotor began to clip the edge of the ship deck and was redirected backward. This, in turn, started to restrict the downward motion of the flow in the rear half of the rotor, which can be clearly seen by comparing the directions of the streamlines and $\Delta\tilde{w}$ in the bottom row between columns

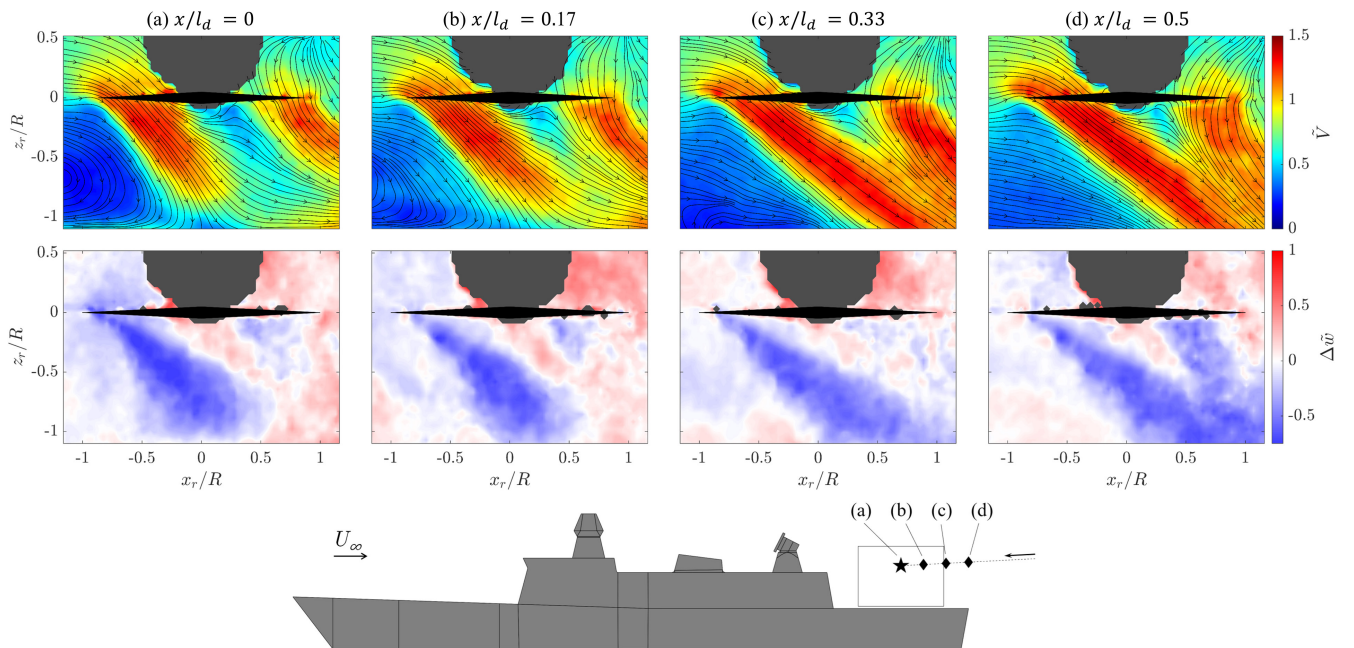


Figure 8: Normalized velocity magnitudes ($\tilde{V}$) and the delta in normalized vertical velocities ($\Delta\tilde{w}$) relative to an isolated rotor out of ground effect, around the approaching rotor within the rotor center plane ($y = 0$) at four locations along the approach under headwind. The black box denotes the field of view for the HOD position marked by the star symbol.

(c) and (d). A significant increase in $\Delta\tilde{w}$ around the rear half of the rotor disk indicates a great reduction in downwash, hence reducing the inflow and increasing local thrust. While the upstream half of the rotor experienced minimal changes, this started to induce a pitch-down moment as previously observed in the load measurements.

As the rotor continued towards the HOD position, this ground effect persisted and was evident in the consistent reduction in downwash velocity on the rear half of the rotor across columns (a), (b), and (c). Additionally, effects from the separated flow region behind the radar and hangar wall began to emerge, appearing as decreasing vertical velocities both above and below the rotor disk in the forward half of the rotor. This further increased the inflow angle on this side of the rotor, reducing local thrust and further enhancing the pitch-down moment as the rotor approached the HOD position.

Next, the flow fields under quartering winds were examined to understand why the rotor, under these conditions, exhibited weaker pitch-down tendencies as it approached the hover position and to assess how the two wind directions differed from each other. Figure 9, again, shows the delta in vertical velocities ($\Delta\tilde{w}$) between an isolated rotor at different locations during the final approach, but under non-zero WOD angles. Streamlines of the actual flow fields are also included for reference.

The top row flow fields in Fig. 9 were obtained under R15 winds, and the bottom row from G15 winds, as indicated by the red and green surrounding boxes, respectively. Both cases showed a noticeable delayed onset of ground-effect-induced pitch-down tendency compared with the headwind condition

shown previously in Fig. 8, indicated by the reduced downwash on the rear half of the rotor. This effect started to occur at location (b), which was further inwards from the stern of the ship compared with the headwind case. Furthermore, between the two wind directions, R15 showed a slightly stronger ground effect that restricted the rear rotor wake more than that observed under G15, as seen by the smaller blue downwash region around the rear-half of the rotor at the HOD position, shown in column (a).

Finally, neither case showed strong effects from separated flow behind the radar and hangar wall compared with the case under headwind, due to the yawed ship shifting the region of strong separated flow. Collectively, due to the delayed ground effect and reduced interactions with the ship separated flow region, the rotor experienced less pitch-down effect from the ship deck under these quartering wind conditions. Additionally, differences in the strength of ground effect resulted in the rotor under R15 winds showing greater changes in pitching moment than under G15 winds during the final approach.

TIME-FREQUENCY ANALYSIS

Apart from the slow-varying trends discussed in the previous section, disturbances from the unsteady ship airwake also inject frequency content into the rotor loading. These fluctuating aerodynamic loads, on both the rotor and the airframe, result in motion that the pilot has to actively counteract in order to maintain stable flight. McRuer (Ref. 24) has shown that frequencies between 0.2 and 2 Hz were typically associated with a high pilot workload, known as the closed-loop pilot response frequency range. Lee and Zan (Ref. 25) later adopted

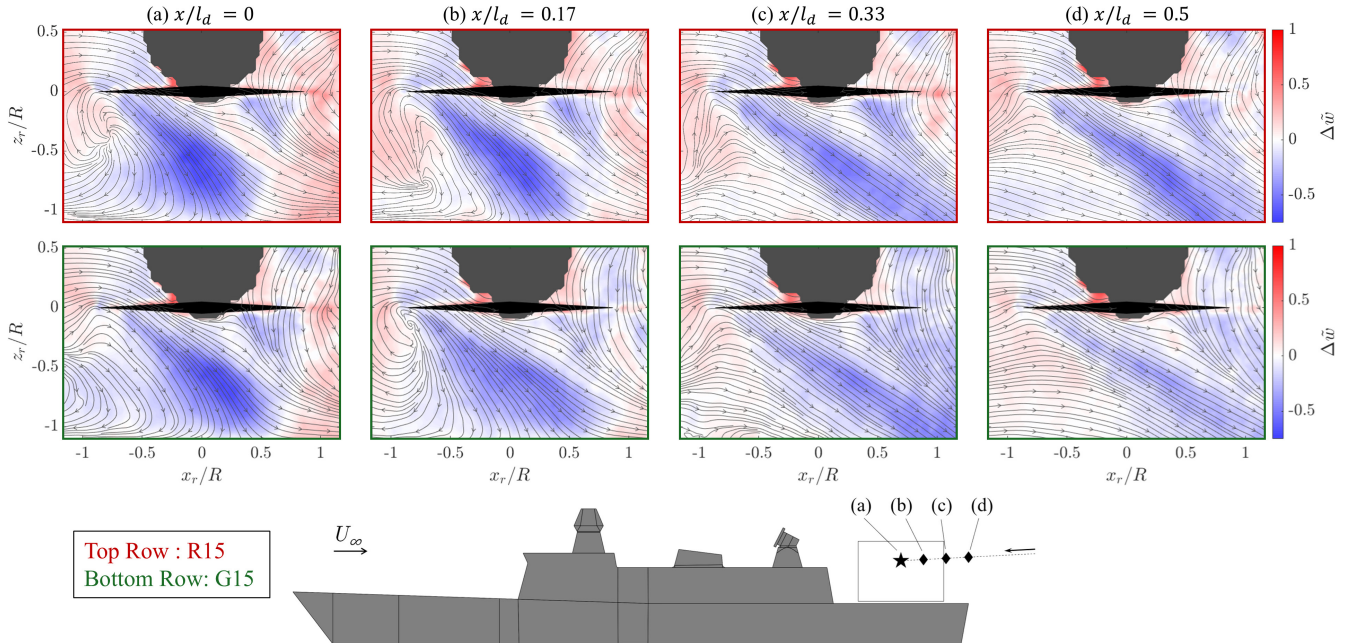


Figure 9: Delta in normalized vertical velocities ($\Delta\tilde{w}$) relative to an isolated rotor out of ground effect, around the approaching rotor within the rotor center plane ($y = 0$) at four locations along the approach under R15 (top row) and G15 (bottom row) WOD. The black box denotes the field of view for the HOD position marked by the star symbol.

this and defined the unsteady root-mean-square (RMS) rotor loads by taking the square root of the time history of loads integrated over the power spectral density (PSD) within the above-mentioned full-scale frequency range. This served as a quantitative metric to assess the unsteadiness and indirectly indicate the associated pilot workload. It has been shown experimentally by McTavish et al. (Ref. 26) that there was a positive correlation between the normalized RMS rotor thrust in subscale experiments and the pilot workload from full-scale flight trials. In subscale experiments, scaling of this frequency range is necessary and was derived in detail by McTavish et al. (Ref. 26). By assuming identical air density, the frequency scale (λ_f), the ratio between frequencies at model and full scale, can be expressed by:

$$\lambda_f = \frac{\lambda_v}{\lambda_D} \quad (1)$$

with λ_v being the velocity scale and λ_D being the length scale. By properly scaling both the rotor thrust coefficient and advance ratio, the frequency can be scaled simply by the ratio of the rotor angular rate of the scaled wind tunnel model to that of the full-scale reference rotorcraft. In the present study, this resulted in a model-scale closed-loop pilot response frequency range of 4.86–48.6 Hz.

In order to assess the relative unsteadiness and pilot workload throughout a continuous dynamic approach towards the ship deck, the continuous wavelet transform (CWT) was used to obtain a time-frequency representation of the evolving loads. The wavelet transform is particularly useful for the analysis of statistically non-stationary and intermittent signals and provides superior frequency resolution at lower frequencies compared with methods such as the short-time Fourier transform (STFT). This is especially important since

the closed-loop pilot response frequency range predominantly includes low-frequency oscillations. In the present study, a Morse wavelet (Refs. 27, 28) that closely resembles the Morlet wavelet was used as the mother wavelet. The CWT works by taking this mother wavelet and sliding it across the target signal in time, checking how well they match at each time instance. This process is performed multiple times while also stretching the mother wavelet to resolve slow features and compressing it to search for fast features. This creates a multiresolution view of the original signal because the large, stretched wavelets capture low-frequency structures over coarser time spans, while small, compressed wavelets capture sharp, high-frequency details over short time spans, effectively analyzing the same signal at different levels of detail in both time and frequency.

Vibrational Effects of Traverse Motion

As mentioned previously, a typical challenge in measurements with dynamic motions comes from the inertial and vibrational loads caused by the motion itself, contaminating the true signals. To quantify and subsequently subtract the effect of the traverse movement, CWT was also performed on the same set of quiescent approach data measured under no freestream velocity and without the rotor spinning. Figure 10 contains the resultant time-dependent magnitude spectra, or scalograms, showing the evolution of frequency contents of the measured rotor thrust component caused by the traverse movement alone and the uncorrected headwind approach. These are ensemble averages over all independent test trials, where N denotes the number of trials performed. The model-scale closed-loop pilot response frequency range is indicated by the region bounded by the two white horizontal lines, which correspond to the

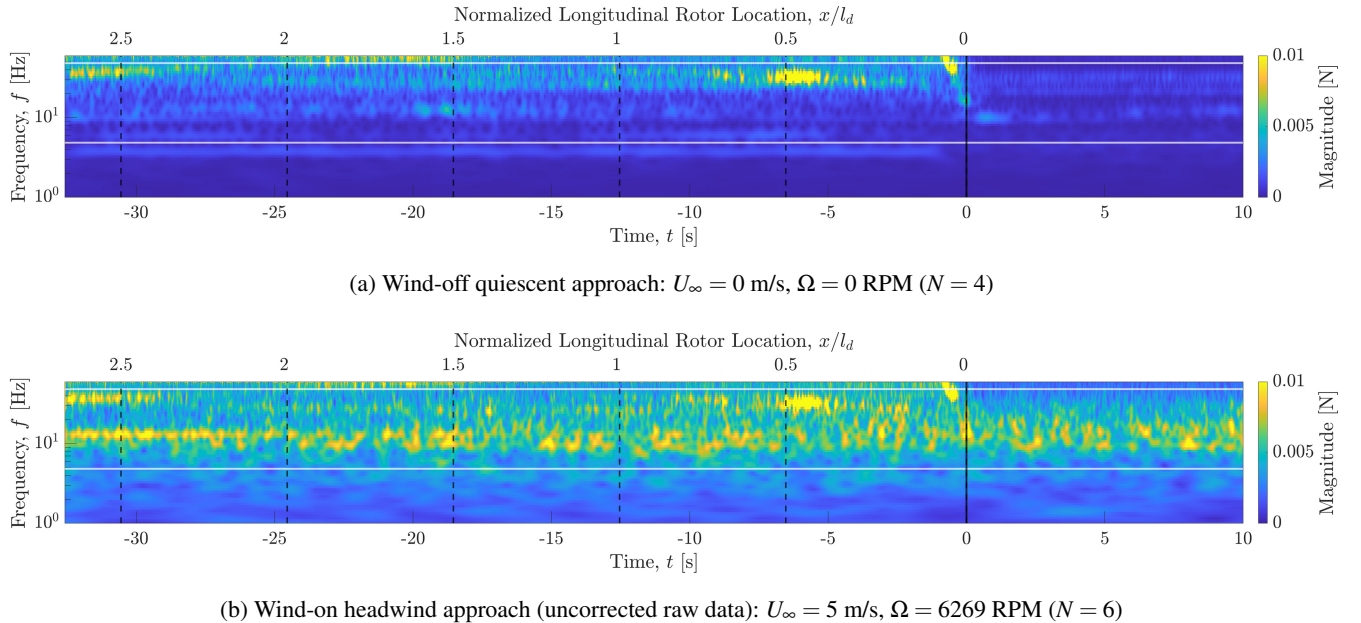


Figure 10: Ensemble averages of wavelet magnitude spectra on rotor thrust during (a) wind-off quiescent approach, and (b) wind-on headwind approach.

upper (48.6 Hz) and lower (4.86 Hz) bounds, respectively. The horizontal axes and labels for time and location in these plots follow the convention used for the rotor loads in Fig. 7.

First, Fig. 10a shows the contributions of the traverse motion on the measured rotor thrust component, obtained from the quiescent approach tests ($U_\infty = 0$ m/s, $\Omega = 0$ RPM). The motion of the traverse system had relatively low signatures within the lower half of the closed-loop pilot response frequency range, particularly under 10 Hz. However, higher magnitudes were seen at higher frequencies, especially above 20 Hz, with intermittent peaks around 35 Hz at $t < -30$ s and -9 s $< t < -3$ s. These peaks were associated with vibrations from the resonance of the long, spinning lead screws at certain lengths. Although there were higher-frequency components associated with the movement, they were well outside the frequency range of interest and were not shown in the graphs. One exception occurred at the end of the motion, where the system gradually slowed down to a stop at $x/l_d = 0$, and one of the higher-frequency peaks slightly crossed the upper bound into the range of interest. After the motion had stopped, there were still small traces of vibrations from the environment and from the structural modes of the traverse system itself.

Next, the CWT result for the rotor operating along a 3 deg glide slope towards the ship deck under headwind (0 deg WOD) before any correction is shown in Fig. 10b. From the figure, it is obvious that the major components from the traverse motion also showed up noticeably in the frequency spectrum. Given that they're of a similar magnitude, or even greater, than the rest of the signals, it is necessary to remove them to prevent misinterpretation of the results. To remove these signals, a direct subtraction was done by essentially subtracting the values in Fig. 10a from those in Fig. 10b. After performing the subtraction, the contributions from the motion of the traverse were greatly reduced, as shown in Fig. 11, and the resultant graph provides a more truthful representation of the aerodynamic impacts on the rotor from the ship airwake. As a result, all subsequent CWT results shown in this study were processed with the same procedure.

Approach under Headwind

Again, looking at Fig. 11, which shows the time history of frequency content in the rotor thrust during the headwind approach, it is noticeable that the high-magnitude frequency

content was primarily concentrated in the 8–20 Hz band. This frequency range equates to the ship width-based Strouhal number (St) of 0.35–0.88. This indicates that the aerodynamic effects of the ship airwake on the unsteadiness of rotor thrust were predominantly narrowband. The width of this high-fluctuation band also changed in width as the rotor moved along the approach path, with a much more concentrated band peaking around 12.5 Hz ($St = 0.55$) at the early stage of the approach ($x/l_d > 2$). As the rotor continued to move closer to the ship ($2 > x/l_d > 1$), intermittent traces of frequency content around 10 Hz ($St = 0.44$) began to emerge, while the occurrence of the original dominant 12.5 Hz signal became increasingly sparse and irregular. Once the rotor entered the final approach ($x/l_d < 1$) and while hovering over the landing deck, the proximity to the ship superstructure resulted in a relatively more energetic and chaotic interaction with the recently shed vortices from the topside features of the ship. These young-age vortices were more structured, less dissipated, and appeared as a relatively wider band of frequency content in the measured rotor thrust.

In order to understand why certain frequencies of the ship airwake were able to remain influential on the unsteady rotor thrust even far downstream of the ship deck, the flow fields around the landing deck region were examined without the presence of the rotor. Stereoscopic PIV was performed separately on the same vertical plane along the ship centerline using the same setup described previously in the methodology section. However, images were acquired at 400 Hz over 4 seconds, with 2 sets of runs, yielding a total of 8 seconds or 3200 flow field snapshots for analysis. A higher sampling rate was used to capture flow features across a wider frequency range within the ship wake. The obtained flow fields were then processed using spectral proper orthogonal decomposition (SPOD), a data-driven technique used to extract coherent flow structures from time-resolved data.

Similar to standard POD, SPOD determines an optimal orthogonal basis that captures the maximum energy content of the data. Its distinction lies in the spatiotemporal nature of the extracted modes, which are orthogonal under a space-time inner product, enabling optimal representation of coherent structures in both space and time. In practice, this is done by splitting the zero-mean, statistically stationary data into overlapping time blocks and Fourier-transforming each block, as in Welch's method. Identical frequencies were then grouped

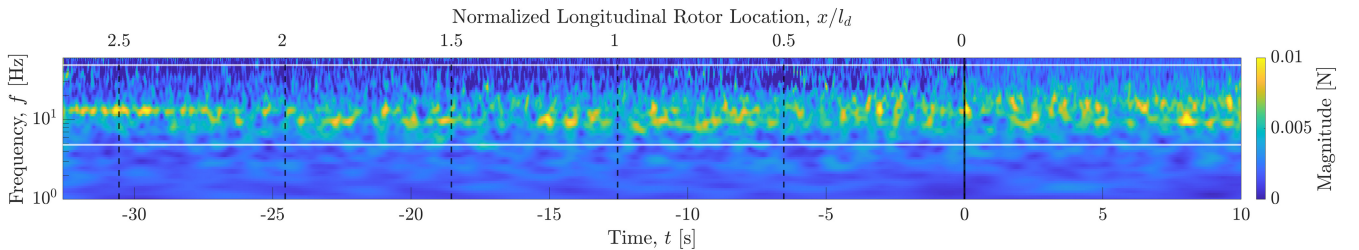


Figure 11: Ensemble averages ($N = 6$) of wavelet magnitude spectra on rotor thrust during a headwind approach after traverse motion contribution was subtracted.

across all blocks, and standard POD was performed on a frequency-by-frequency basis. The SPOD modes are the eigenvectors, and the leading modes identify the most energetic spatial structures that are coherent at a specific oscillation frequency, while the eigenvalues provide the modal energy spectrum. Details of the SPOD method are introduced and elaborated by Sieber et al. (Ref. 29) and also Schmidt and Colonius (Ref. 30). In this present study, a window length of 256 snapshots and 50% overlap between adjacent time blocks were used to perform the SPOD.

Figure 12 shows the spectra of eigenvalues of each mode. The red line indicates the sum of all eigenvalues at each frequency and is analogous to the power spectral density. The black line represents the eigenvalue of the first mode of each frequency, and trailing modes were plotted as thin gray lines with decreasing intensity. The nature of the airwake of the NATO-GD appears to be broadband but is mostly dominated by low frequencies, with similar cumulative energy across frequencies up to 10 Hz ($St = 0.44$). This was also observed in previous simulation studies on the NATO-GD airwakes under identical scaling conditions by Ashok and Rauleder (Ref. 31). Additionally, frequencies below 20 Hz ($St = 0.88$) exhibited more pronounced dominant modes, i.e., greater energy concentrated towards the leading modes.

This is better visualized in Fig. 13, which shows the normalized relative percentage of energy accounted for by individual modes as a function of frequency. Frequency components lower than 20 Hz ($St = 0.88$) had their first mode containing at least 15% of the total energy at each frequency, with a peak over 25% occurring around 7.8 Hz ($St = 0.34$). This suggests that these low-frequency modes, especially within 5–20 Hz, exhibit more prominent coherent structures. This frequency band aligns with the previous observation from the CWT results on unsteady rotor loads (Fig. 11). These low frequencies are highly energetic, and associated coherent structures are likely to be carried downstream more than the higher-frequency counterparts. Higher-frequency content had the energy more evenly distributed across all modes, as seen by the more uniform color in Fig. 13, signifying more random and chaotic flow features. However, from Fig. 11, only a few

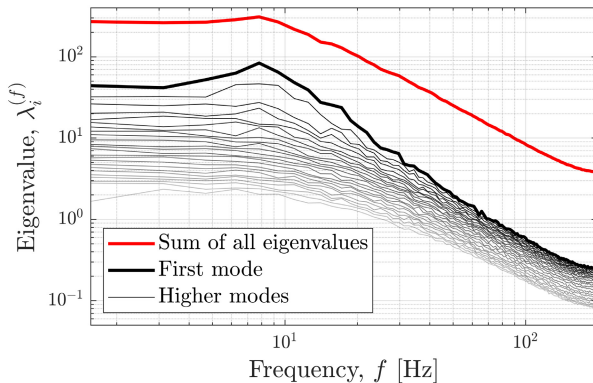


Figure 12: Eigenvalues of the SPOD modes of the flow field above the landing deck of a standalone NATO-GD.

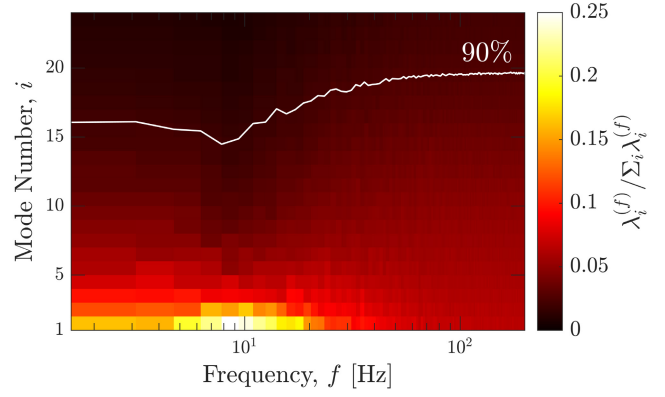


Figure 13: Percentage of energy accounted for by each mode as a function of frequency. Solid white line indicates number of modes required to preserve 90% of energy at each frequency.

dominant frequencies were experienced by the rotor furthest downstream of the ship, suggesting that specific flow features, or the location of these features, might have had an effect.

To investigate the flow structures in the ship airwake associated with specific frequencies, the normalized velocity components from the first spatial mode at $f = 7.8, 9.4, 10.9$, and 12.5 Hz were analyzed, as shown in Fig. 14. Note that the magnitudes of these components were scaled by the eigenvalues of each mode. The dominant SPOD modes at these frequencies exhibit distinct spatial characteristics and show prominent features in different velocity components, each originating from different regions of the ship's superstructure.

First, by examining Fig. 14a, which is the first mode associated with $f = 7.8$ Hz, this mode appears to have been dominated by the large-scale lateral oscillations from the bulky superstructure, including the hangar and the radar above it. According to the eigenvalues shown in Figs. 12 and 13, this is the most energetic mode across the entire frequency spectrum. However, despite the high energy content, this frequency mode was not prominent in the rotor thrust during the initial approach, likely because the oscillating fluid motion was predominantly in the lateral direction. The lack of significant vertical motion indicates that there would be minimal perturbation on the rotor inflow. Additionally, the vertical location of this fluid structure was relatively low, with the highest intensity just below the hangar wall, at $z \approx 0.06$ m above the deck, well below the rotor position of $z = 0.137$ m at the beginning of the approach. Thus, even if this structure were convected downstream, it is expected to have relatively little influence on the rotor performance further downstream.

Stepping up to the next frequency of 9.4 Hz, shown in Fig. 14b, the flow feature seems to still be dominated by lateral oscillations. However, the source of this fluctuation shifted upward ($z \approx 0.1$ m) and became more concentrated towards the body of the radar on top of the superstructure. Despite also being predominantly a lateral flow feature, since this location was directly upstream of the rotor during hover and the final approach, intermittent signatures around this frequency were

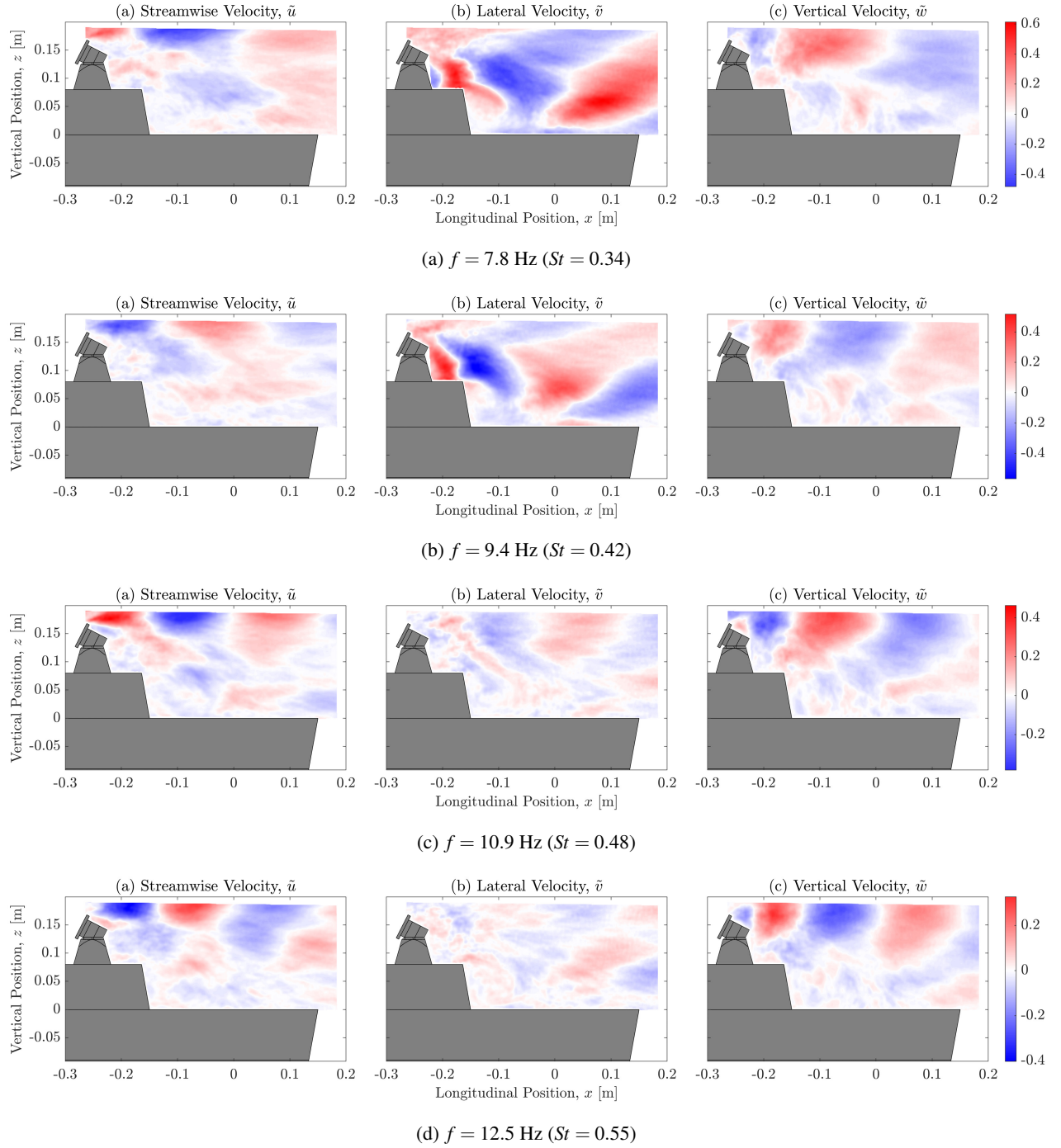


Figure 14: First SPOD modes of isolated ship with normalized velocity components associated with several frequencies.

also observed in Fig. 11, increasing fluctuations in the rotor thrust and, in turn, the pilot workload.

Finally, looking at the even higher frequencies at 10.9 and 12.5 Hz, shown in Figs. 14c and 14d respectively. These frequencies were within the range observed to dominate the unsteady rotor thrust further downstream of the approach. Flow features show that these higher-frequency modes were primarily structures originating from flow separation at the top-side edge of the radar dish, with very little lateral motion. The periodic nature of the spatial modes along the

longitudinal direction, both in the streamwise and vertical velocity components, indicates that these were strong coherent vortical structures lying within the x - z plane. These shed vortices were transported downstream, thereby influencing the rotor inflow through the oscillating vertical velocity component and introducing periodic fluctuations in rotor thrust at these specific frequencies. Furthermore, the fact that the prominent flow structures were higher in the vertical direction, which better aligns with the rotor position at earlier approach stages, resulted in the vortical wakes directly interacting with the rotor

inflow, enhancing the unsteady perturbations in rotor loads.

In summary, during the early approach segment, aerodynamic interactions were dominated by periodic vortical structures shed over the top of the radar dish. As the rotor continued to close in towards the ship deck along a 3 deg glide slope, it began to experience lower-frequency effects from geometric features at lower elevations, such as the radar base and the hangar. Finally, during the final approach and hover above the landing deck, the proximity of the rotor to the ship superstructure caused it to interact with various flow structures with different characteristic frequencies, which appeared as relatively broadband frequency signatures in the rotor thrust compared with the earlier approach stages.

Relative-Wind Approaches under Quartering Winds

As the WOD angle changes, the specific characteristics and locations of the airwake structures are expected to shift relative to the rotor and, in turn, alter the unsteady aerodynamic interactions. Figure 15 shows the CWT results of the rotor thrust along relative-wind approaches under different WOD angles, specifically G15 (starboard-side wind) and R15 (port-side wind). Again, in these two cases, the rotor approached the ship deck along the exact same trajectory as the headwind approach, a 3-deg glide slope directly moving upstream, while the heading of the ship was changed relative to the freestream direction. Thus, the plots shown in Fig. 15 already had the same movement noise results from the headwind approach, previously shown in Fig. 10a, subtracted from the CWT results for both cases.

From Fig. 15a, the level of unsteady fluctuations in rotor thrust was similar to that observed during a headwind, especially when near the landing deck and while hovering above the deck

($x/l_d < 1$). These unsteady components also occurred within a similar frequency band, indicating that similar flow features originating from the ship superstructure at these frequencies were still dominant even though the relative wind was coming from the starboard side of the ship. This oblique wind direction shifted the unsteady ship airwake slightly towards the retreating side of the rotor. As previously reported by Chen and Rauleder (Ref. 15), the interaction between the ship airwake and the retreating side was the primary source of the unsteady fluctuations in thrust in static hover tests of a generic rotor within ship airwakes. However, further downstream during the initial approach, there was little to no signature caused by the 12.5 Hz vortices shed from the top of the radar disk as seen under a headwind. This was likely due to these flow structures being less energetic after traveling downstream or after being convected farther laterally and vertically by the low-pressure ship airwake on the leeward side of the ship, resulting in minimal interactions with the rotor.

On the other hand, under a R15 WOD angle, the ship airwake was shifted towards the advancing side of the rotor, with the majority of it missing the retreating side of the rotor. As seen in Fig. 15b, this resulted in a significant reduction in unsteadiness throughout the entire approach trajectory within the closed-loop pilot response frequency range compared with both the headwind and G15 WOD cases. Combined with a lower pitch-down tendency near final approach, it seems that a relative-wind approach under R15 WOD could result in the least pilot workload among all three flight conditions examined based on the above observation in the present study. However, the full-scale pilot workload under shipboard operations is dominated not only by these aerodynamic interactions, but also encompasses factors such as the available visual cues, ship motion, weather conditions, etc. The compounded effects from

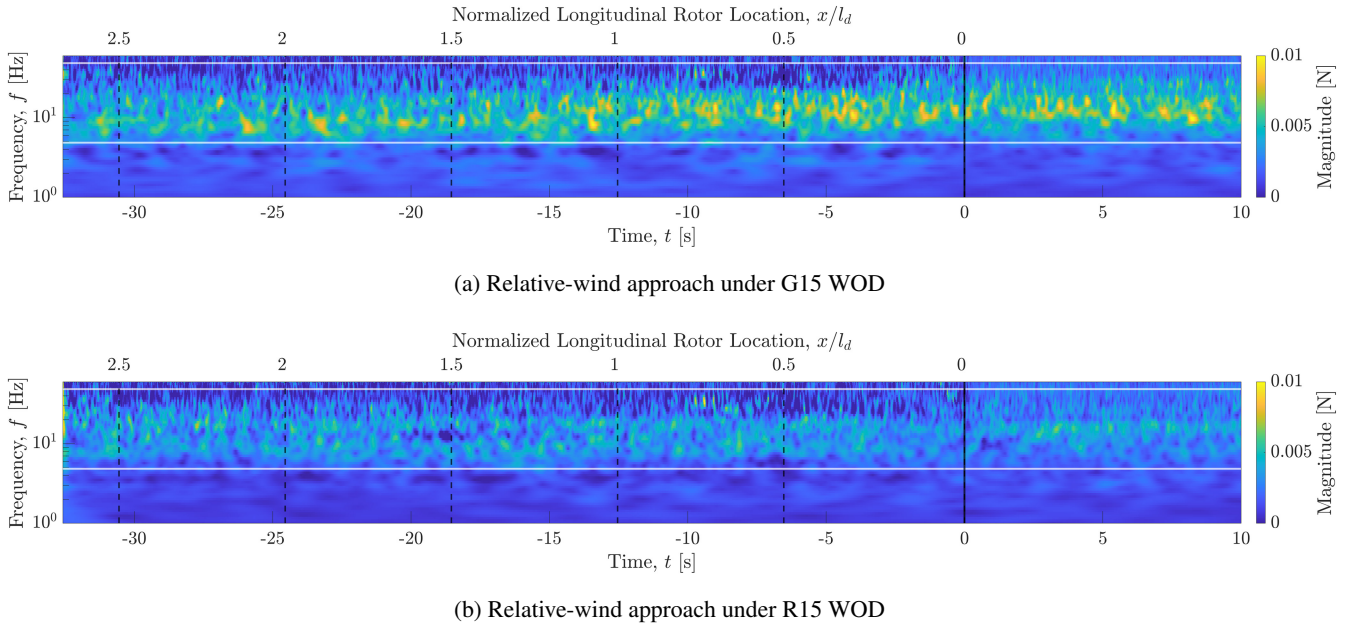


Figure 15: Ensemble averages ($N = 6$) of wavelet magnitude spectra on rotor thrust during a relative-wind approach path under quartering winds.

these factors further complicate the ship-rotor DI problem and underscore the need for more comprehensive research efforts to ensure safe and efficient operations.

CONCLUSIONS

The dynamic approach experiments in this study revealed several key features of the ship-rotor interaction during a recovery process of a representative rotor towards the landing deck of a representative model-scale destroyer:

1. The rotor thrust generally changed as a function of proximity to the ship, with the direction and magnitude of the variation depending strongly on the WOD angle.
2. Under headwind conditions, the rotor encountered increasingly strong pitch-down tendencies as it approached the deck, resulting from the combined influence of ground effect and the separated recirculating ship airwake near the hangar region.
3. Quartering winds altered the location of the ship airwake relative to the rotor disk and caused changes in the strength of ground effect, reducing the pitch-down trend compared with the headwind case, also producing different thrust trends for the G15 and R15 WOD cases.
4. Rolling moments varied less than thrust and pitching moment, suggesting partial cancellation of asymmetric inflow effects laterally across the rotor disk during the approach.
5. Time-frequency analysis showed that the rotor response was dominated by narrowband unsteady content in the closed-loop pilot response frequency range, increasing the pilot workload, with the strongest energetic features occurring at specific frequencies associated with coherent ship-topside flow structures.
6. SPOD results indicated that the most influential unsteady flow structures originated from flow separation over the radar and other topside features, and that these flow structures evolved downstream to interact with the rotor at different stages of the approach.

Overall, the results show that the dynamic approach captured both slow- and fast-changing transient aerodynamic effects with fine spatial resolution. These evolving trends in rotor loading, particularly the wind-angle-dependent thrust changes and pronounced pitch-down tendencies near the deck, will require pilots to shift control inputs and increase collective/cyclic effort to maintain stable flight. The narrowband unsteady fluctuations concentrated within the closed-loop pilot response frequency range would further degrade handling qualities by inducing persistent low-frequency oscillations that demand active control corrections. Collectively, these dynamic effects highlight challenging aerodynamic interactions during rotorcraft approaches to ship decks or similar complex environments,

emphasizing the need for novel techniques that account for dynamical effects to refine estimates of pilot workload and ship-helicopter operational limits.

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Author Contacts

Wei-Han Chen: wchen669@gatech.edu

Dounya Jarrad: djarrad3@gatech.edu

Juergen Rauleder: Juergen.Rauleder@gatech.edu

REFERENCES

1. Wilkinson, C. H., Zan, S. J., Gilbert, N. E., and Funk, J. D., "Modelling and Simulation of Ship Air Wakes for Helicopter Operations: A Collaborative Venture," NATO RTO-MP-15, Proceedings of the RTO AVT Symposium on Fluid Dynamics Problems of Vehicles Operating Near or in the Air-Sea Interface, October 1998.
2. Zan, S. J., "Surface Flow Topology for a Simple Frigate Shape," *Canadian Aeronautics and Space Journal*, Vol. 47, NRC Report No. AL-2001-0114, NPARC No. 8928559, March 2001.
3. Wall, A., Lee, R., Barber, H., and McTaggart, K., "The NATO Generic Destroyer—A Shared Geometry for Collaborative Research into Modelling and Simulation of Shipboard Launch and Recovery," *Open Science Canada*, 2020.
4. Seth, D., Leishman, J. G., Gnanamanickam, E., and Zhang, Z., "Time-Resolved Ship Airwake Measurements in a Simulated Atmospheric Boundary Layer," *Journal of Aircraft*, Vol. 58, (3), May 2021, pp. 624–649. DOI: 10.2514/1.C035886.
5. Zhu, N., Zhang, Z., Gnanamanickam, E., and Leishman, J. G., "Space-Time Characterization of Ship Airwakes," *AIAA Journal*, Vol. 61, (2), February 2023, pp. 681–697. DOI: 10.2514/1.J062093.
6. Palm, K. H., Mazzilli, G. A., Zhang, Z., Gnanamanickam, E. P., and Leishman, J. G., "Characterization of Flow Structures Within the Airwake of NATO Generic Destroyer," AIAA SciTech 2025 Forum, 2025. DOI: 10.2514/6.2025-1077.
7. Yon, S., and Li, S. K., "Longitudinal Gust Effects on Ship Airwake Characteristics for Helicopter Landing Operations with Correlations to Pilot

- Workload Factor,” AIAA SCITECH 2026 Forum, 2026. DOI: 10.2514/6.2026-1054.
8. Zan, S. J., “Experimental Determination of Rotor Thrust in a Ship Airwake,” *Journal of the American Helicopter Society*, Vol. 47, (2), April 2002, pp. 100–108. DOI: 10.4050/JAHS.47.100.
 9. Wang, Y., Curran, J., Padfield, G., and Owen, I., “AirDyn: an instrumented model-scale helicopter for measuring unsteady aerodynamic loading in airwakes,” *Measurement Science and Technology*, Vol. 22, (4), 045901, 2011. DOI: 10.1088/0957-0233/22/4/045901.
 10. Kääriä, C. H., Wang, Y., White, M. D., and Owen, I., “An experimental technique for evaluating the aerodynamic impact of ship superstructures on helicopter operations,” *Ocean Engineering*, Vol. 61, 2013, pp. 97–108. DOI: 10.1016/j.oceaneng.2012.12.052.
 11. Taymourtash, N., Zagaglia, D., Zanotti, A., Muscarello, V., Gibertini, G., and Quaranta, G., “Experimental study of a helicopter model in shipboard operations,” *Aerospace Science and Technology*, Vol. 115, 106774, 2021. DOI: 10.1016/j.ast.2021.106774.
 12. Matías, J., Bardera, R., Franchini, S., Barroso, E., and Sor, S., “A comparative analysis of helicopter recovery maneuvers on a SFS by means of PIV and balance measurements,” *Ocean Engineering*, Vol. 275, May 2023, pp. 114119. DOI: 10.1016/j.oceaneng.2023.114119.
 13. Silva, M., Schatzman, D., and Ramasamy, M., “Sub-Scale Investigation of Coupled Rotor/Upwind Ship Aerodynamic Interactions,” Proceedings of the Vertical Flight Society’s 6th Decennial Aeromechanics Specialists’ Conference, 2024.
 14. Taymourtash, N., Zanotti, A., Gibertini, G., and Quaranta, G., “Simulation and testing of helicopter-ship aerodynamic interaction,” 47th European Rotorcraft Forum (ERF 2021), 2021.
 15. Chen, W. H., and Rauleder, J., “Unsteady Loads of a Scaled Rotor on and near the Landing Deck of the NATO Generic Destroyer with Concurrent Stereo PIV and Surface Pressure Measurements,” *Journal of the American Helicopter Society*, Vol. 69, (4), 2024. DOI: 10.4050/JAHS.69.042008.
 16. Ashok, S. G., and Rauleder, J., “Real-Time Coupled Ship-Rotorcraft Interactional Simulations Using GPU-Accelerated Lattice-Boltzmann Method,” *Journal of the American Helicopter Society*, Vol. 70, (2), 2025, pp. 1–25. DOI: 10.4050/JAHS.70.022005.
 17. Semelka, A., and Rauleder, J., “Experimental Investigation of Propeller–Wing Interactions during Dynamic Tiltrotor Conversion Maneuver,” Vertical Flight Society’s 82nd Annual Forum Technology Display, 2026.
 18. Chen, W. H., and Rauleder, J., “Wind Tunnel Investigation of a Rotor Under the Influence of Unsteady Ship Airwakes with Azimuth-Related Measurements,” Proceedings of the 81st Annual Forum of the Vertical Flight Society, Virginia Beach, VA, USA, 05 2025. DOI: 10.4050/F-0081-2025-47.
 19. NATO AVT-315 Task Group, “Comparative Assessment of Modelling and Simulation Methods of Shipboard Launch and Recovery of Helicopters,” Technical Report STO-TR-AVT-315, 2023. DOI: 10.14339/STO-TR-AVT-315.
 20. Healey, J. V., “Establishing a database for flight in the wakes of structures,” *Journal of Aircraft*, Vol. 29, (4), 1992, pp. 559–564. DOI: 10.2514/3.46202.
 21. Rosenfeld, N., Kimmel, K., and Sydney, A. J., “Investigation of ship topside modeling practices for wind tunnel experiments,” Proceedings of the 53rd AIAA Aerospace Sciences Meeting, Kissimmee, FL, January 5–9, 2015, 0245. DOI: 10.2514/6.2015-0245.
 22. Westerweel, J., and Scarano, F., “Universal outlier detection for PIV data,” *Experiments in fluids*, Vol. 39, (6), 2005, pp. 1096–1100. DOI: 10.1007/s00348-005-0016-6.
 23. Curtiss, H., Sun, M., Putman, W., and Hanker, E., “Rotor aerodynamics in ground effect at low advance ratios,” *Journal of the American Helicopter Society*, Vol. 29, (1), 1984, pp. 48–55. DOI: 10.4050/JAHS.29.48.
 24. McRuer, D. T., “Interdisciplinary interactions and dynamic systems integration,” *International Journal of Control*, Vol. 59, (1), 1994, pp. 3–12. DOI: 10.1080/00207179408923067.
 25. Lee, R. G., and Zan, S. J., “Wind tunnel testing of a helicopter fuselage and rotor in a ship airwake,” *Journal of the American Helicopter Society*, Vol. 50, (4), 2005, pp. 326–337. DOI: 10.4050/1.3092869.
 26. McTavish, S., Wall, A., and Lee, R., “A Methodology to Correlate Simulated Airwake Data and Unsteady Helicopter Load Measurements to Shipboard Helicopter Flight Test Data,” Proceedings of the 14th International Conference on Wind Engineering, 2015.
 27. Olhede, S. C., and Walden, A. T., “Generalized morse wavelets,” *IEEE Transactions on Signal Processing*, Vol. 50, (11), 2002, pp. 2661–2670. DOI: 10.1109/TSP.2002.804066.
 28. Lilly, J. M., and Olhede, S. C., “Generalized Morse wavelets as a superfamily of analytic wavelets,” *IEEE Transactions on Signal Processing*, Vol. 60, (11), 2012, pp. 6036–6041. DOI: 10.1109/TSP.2012.2210890.
 29. Sieber, M., Paschereit, C. O., and Oberleithner, K., “Spectral proper orthogonal decomposition,” *Journal of Fluid Mechanics*, Vol. 792, 2016, pp. 798–828. DOI: 10.1017/jfm.2016.103.
 30. Schmidt, O. T., and Colonius, T., “Guide to spectral proper orthogonal decomposition,” *Aiaa journal*, Vol. 58, (3), 2020, pp. 1023–1033. DOI: 10.2514/1.J058809.
 31. Ashok, S. G., and Rauleder, J., “Mid-fidelity NATO generic destroyer static and moving-ship airwake simulations using the Lattice-Boltzmann method compared with experimental data,” *Ocean Engineering*, Vol. 298, 2024, pp. 117264. DOI: 10.1016/j.oceaneng.2024.117264.